

ANIRUDDH KULKARNI

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PROFESSIONAL SUMMARY

Software Engineer with 4.5+ years of experience developing real-time control software for semiconductor equipment. Skilled in C/C++, Java, and VxWorks, with expertise in Etch chamber control, process diagnostics, and field support. Delivered baseline releases, optimized system performance, and supported advanced manufacturing processes.

EXPERIENCE

UST — Software Developer II (Client: Applied Materials) Bengaluru, India

Jan 2024 – Present

Etch Chamber Control, VxWorks RTOS, Embedded Systems

- Engineered real-time Etch chamber control software in C/Java on VxWorks RTOS, improving reliability and responsiveness.
- Developed and tuned control algorithms to stabilize plasma behavior and enhance repeatability across chamber variants.
- Delivered baseline releases, resolved field issues via root-cause analysis and on-tool debugging, collaborating with cross-functional teams.

Spanidea Systems — Associate Software Engineer Bengaluru, India

Sept 2022 – Dec 2023

5G PHY Channel Processing, Xilinx FPGA, Signal Optimization

- Designed and implemented 4G/5G Layer-1 PHY channel processing modules in C++/Matlab on Xilinx FPGA.
- Optimized latency and resource utilization using Vitis & Vivado for high-performance signal processing.
- Applied 3GPP RF and protocol standards to ensure compliance and robust PHY/MAC layer integration.

Veoneer — Intern Bengaluru, India

Oct 2021 – Aug 2022

mmWave Radar, Signal Processing, Lane Change Assist

- Developed mmWave radar signal-processing pipeline for Lane Change Assist (LCA) use cases in C and Matlab.
- Validated algorithms against ISO-17387 datasets, improving lane-change detection accuracy.
- Contributed to systems engineering practices to ensure reliable radar solution development.

SKILLS

- **Programming and Development:** C, C++, Java, VxWorks RTOS
- **Domain Knowledge:** Semiconductor Etch Process Control, Embedded Systems, Software Debugging
- **AI & Automation:** Leveraged GitHub Copilot and Claude CLI to accelerate debugging, code generation, and workflow automation.

EDUCATION

- **Master of Communication Systems**
RV College of Engineering, Bengaluru; *CGPA* — 8.68 2020–2022
- **Bachelor of Electronics and Communication Systems**
Dr. AIT College, Bengaluru; *CGPA* — 8.17 2015–2019